

Discrete Mathematics & Number Theory: The Master Guide

This guide covers the core concepts of Relations, Functions, Ordered Sets, Number Theory, and Algebraic Structures. Each section includes deep theoretical explanations, simplified analogies for beginners, and step-by-step problem-solving strategies.

Unit 1: Relations & Functions

1. Introduction to Relations & Types of Relations

Deep Concept

In mathematics, a **Relation** is a way of describing a connection between elements of sets. Formally, if we have two sets A and B , a relation R is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we say " a is related to b ".

Explain Like I'm 5

Imagine a classroom. The set A is all the students. A relation is a specific rule, like "is friends with."

- If John is friends with Mary, they are "related" in this context.
- If John doesn't know Steve, they are not "related." A relation is just a list of all the pairs of people who follow that rule.

Sub-Concepts (Types of Relations)

1. **Reflexive:** Every element is related to itself (aRa for all a).
2. **Symmetric:** If a is related to b , then b is related to a ($aRb \implies bRa$).

3. **Anti-Symmetric:** If aRb and bRa , then a must equal b (No mutual relationship unless it's the same person).
4. **Transitive:** If aRb and bRc , then aRc .

Problem Solving: Identifying Relation Types

Problem: Let $A = \{1, 2, 3\}$. $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$. Determine the type.

How to Solve:

1. **Check Reflexive:** Look for $(1, 1)$, $(2, 2)$, $(3, 3)$. Are they all there? Yes. It is Reflexive.
2. **Check Symmetric:** Look for pairs where $a \neq b$. We have $(1, 2)$. Do we have $(2, 1)$? No. It is NOT Symmetric.
3. **Check Transitive:** Check "chains". $(1, 1)$ leads to $(1, 2) \rightarrow (1, 2)$. This holds. It is Transitive.

2. Properties of Relations: Equivalence & Partial Orders

Deep Concept

We classify relations based on the combination of properties they possess.

- **Equivalence Relation:** Reflexive + Symmetric + Transitive. This partitions a set into "classes" where everything inside is "equal" in some way.
- **Partial Order Relation:** Reflexive + Anti-Symmetric + Transitive. This arranges items in a hierarchy (ordering).



Explain Like I'm 5

- **Equivalence:** Think of sorting Lego bricks by color. All red bricks go in one bucket. A red 2×4 brick is "equivalent" to a red 1×1 brick because they are both red. They live in the same "class."

- **Partial Order:** Think of a family tree. Grandfather $\rightarrow$ Father $\rightarrow$ Son. You can't be your own father (Anti-symmetric), and if Grandpa is older than Dad, and Dad is older than Son, Grandpa is older than Son (Transitive).

Problem Solving: Proving Equivalence

Problem: Let R be a relation on Integers defined by xRy if $x - y$ is even. Is this an Equivalence relation?

How to Solve:

1. **Test Reflexive:** Is $x - x$ even? 0 is even. Yes.
2. **Test Symmetric:** If $x - y$ is even, is $y - x$ even? If $x - y = 2k$, then $y - x = -2k$. Yes.
3. **Test Transitive:** If $x - y = 2a$ and $y - z = 2b$, add them: $(x - y) + (y - z) = x - z = 2(a + b)$. Since $2(a + b)$ is even, Yes. **Conclusion:** It is an Equivalence Relation.

3. Functions – Types, Composition, Inverses

Deep Concept

A function is a special relation where every input (domain) goes to exactly one output (codomain).

- **One-to-One (Injective):** No two inputs give the same output.
- **Onto (Surjective):** Every possible output is used at least once.
- **Bijective:** Both One-to-One and Onto. Only bijective functions have inverses.

Explain Like I'm 5

Think of a Vending Machine.

- **Function:** You press a button ($A1$), you get a snack. You never press $A1$ and get a chip *and* a candy.

- **One-to-One:** $A1$ gives chips, $B1$ gives candy. Different buttons $\rightarrow$ different snacks.
- **Onto:** The machine is empty at the end; every snack slot was linked to a button.
- **Inverse:** Putting the snack back in to get your money back (reversing the process).

Problem Solving: Composition & Inverse

Problem: $f(x) = 2x + 3$. Find $f^{-1}(x)$.

How to Solve:

1. Replace $f(x)$ with y : $y = 2x + 3$.
2. Swap x and y : $x = 2y + 3$.
3. Solve for y :
 - $x - 3 = 2y$
 - $y = \frac{x-3}{2}$
4. **Result:** $f^{-1}(x) = \frac{x-3}{2}$.

4. POSET (Partially Ordered Sets) and Lattices

Deep Concept

- **POSET:** A set equipped with a partial order relation (Reflexive, Anti-symmetric, Transitive). We visualize these using **Hasse Diagrams**.
- **Lattice:** A specific type of POSET where *every pair* of elements has a Least Upper Bound (LUB/Join) and a Greatest Lower Bound (GLB/Meet).

Explain Like I'm 5

- **POSET:** A corporate ladder. The CEO is at the top, interns at the bottom. Some people report to managers, but two managers might not report to each other (they are incomparable).
- **Lattice:** A perfectly organized chart where any two people have a specific common boss (LUB) and a specific common subordinate (GLB).

Problem Solving: Hasse Diagrams

Problem: Draw the Hasse diagram for divisibility on $A = \{1, 2, 3, 6\}$.

How to Solve:

1. **List relations:** 1 divides everything. 2 divides 6. 3 divides 6.
2. **Draw nodes:** Put numbers that divide others lower down.
3. **Connect lines:**
 - 1 is at the bottom.
 - Draw lines from 1 to 2 and 1 to 3.
 - Draw lines from 2 to 6 and 3 to 6.
 - 6 is at the top.
4. **Result:** A diamond shape.

Unit 2: Number Theory

5. Modular Arithmetic – Basics, Congruence

Deep Concept

Modular arithmetic is a system of arithmetic for integers, where numbers "wrap around" after reaching a certain value (the modulus). Notation: $a \equiv b \pmod{n}$ means a and b have the same remainder when divided by n . Equivalently, n divides $(a - b)$.

Explain Like I'm 5

This is "Clock Math." If it is 10:00 and you add 5 hours, on a standard number line, it is 15. But on a clock, it is 3:00. Why? Because the clock "wraps around" every 12 hours. $15 \equiv 3 \pmod{12}$.

Problem Solving: Finding Remainders

Problem: What is $15 + 27 \pmod{4}$?

How to Solve:

1. Find remainder of 15 divided by 4: $15 = 4 \times 3 + 3$. Remainder is **3**.
2. Find remainder of 27 divided by 4: $27 = 4 \times 6 + 3$. Remainder is **3**.
3. Add remainders: $3 + 3 = 6$.
4. Find remainder of 6 divided by 4: $6 = 4 \times 1 + 2$.
5. **Answer:** 2.

6. Euclid's Algorithm & Multiplicative Inverse

Deep Concept

- **Euclid's Algorithm:** An efficient method to find the Greatest Common Divisor (GCD) of two numbers by repeatedly applying the modulo operation.

- **Multiplicative Inverse:** Finding an x such that $ax \equiv 1 \pmod{n}$. This is essential for cryptography (RSA).

Explain Like I'm 5

- **GCD:** You have a rectangular chocolate bar of size 30×21 . You want to cut it into the largest possible equal squares with no leftovers. Euclid's algorithm helps you calculate the size of that square.
- **Inverse:** Regular math says $5 \times \frac{1}{5} = 1$. In modular math, we don't have fractions. We need a whole number that acts like $\frac{1}{5}$.

Problem Solving: Euclid's Algorithm

Problem: Find $\text{GCD}(105, 252)$.

How to Solve:

1. Divide larger by smaller: $252 = 105 \times 2 + 42$ (Remainder 42).
2. Shift left (Old Divisor becomes new Dividend): $105 = 42 \times 2 + 21$ (Remainder 21).
3. Shift left: $42 = 21 \times 2 + 0$ (Remainder 0).
4. **Stop.** The last non-zero remainder is the GCD.
5. **Answer:** 21.

Unit 3: Recurrence & Algebra

7. Recurrence Relations – Definition & Examples

Deep Concept

A recurrence relation is an equation that defines a sequence recursively; each term is defined as a function of the preceding terms. Example: Fibonacci Sequence $F_n = F_{n-1} + F_{n-2}$.

Explain Like I'm 5

Imagine you are climbing stairs. To reach step 10, you must have come from either step 9 (taking a small step) or step 8 (taking a double step). So, [Ways to step 10] = [Ways to step 9] + [Ways to step 8]. You define where you are based on where you just were.

Problem Solving: Modeling

Problem: Determine the recurrence relation for the number of bacteria if the population doubles every hour.

How to Solve:

1. Let a_n be population at hour n .
2. Logic: Population now = 2 times Population 1 hour ago.
3. **Relation:** $a_n = 2a_{n-1}$.

8. Solving Recurrence Relations

Deep Concept

We usually solve **Linear Homogeneous Recurrence Relations with Constant Coefficients** (like $a_n = 5a_{n-1} - 6a_{n-2}$).

Method:

1. Turn into Characteristic Equation ($r^2 - 5r + 6 = 0$).
2. Find roots (r_1, r_2).
3. General Solution: $a_n = C_1(r_1)^n + C_2(r_2)^n$.

Explain Like I'm 5

Solving a recurrence relation is like finding the "DNA" of the sequence. Instead of adding up the previous numbers every single time to get to the 100th number, we find a "Master Formula" that lets us jump straight to the answer.

Problem Solving: Step-by-Step

Problem: Solve $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 1, a_1 = 4$.

How to Solve:

1. **Characteristic Equation:** Replace a_n with r^2 , a_{n-1} with r , etc. $r^2 - 5r + 6 = 0$.
2. **Factor:** $(r - 2)(r - 3) = 0$. Roots are $r = 2, 3$.
3. **General Form:** $a_n = c_1(2)^n + c_2(3)^n$.
4. **Find Constants using initial values:**
 - For $n = 0$: $c_1 + c_2 = 1$.
 - For $n = 1$: $2c_1 + 3c_2 = 4$.
5. **Solve System:** $c_1 = -1, c_2 = 2$.
6. **Final Answer:** $a_n = -1(2)^n + 2(3)^n$.

10. Groups, Rings & Fields

Deep Concept

These are Algebraic Structures (sets with operations like $+$ and $\times$).

1. **Group:** A set with 1 operation (usually $+$ or $\times$). Must have Closure, Associativity, Identity, and Inverse.

2. **Ring:** A set with 2 operations $(+, \times)$. A Group under $+$, but only associative under $\times$ (and distributive).
3. **Field:** A Ring where division is possible (except by zero).

Explain Like I'm 5

Imagine different board games have different rule books.

- **Group:** Basic Checkers. You can move forward and backward (Inverse), and stay still (Identity).
- **Ring:** Integers $(\dots - 2, -1, 0, 1, 2 \dots)$. You can add, subtract, and multiply. But you can't always divide (1 divided by 2 isn't an integer).
- **Field:** Real Numbers. You can add, subtract, multiply, and divide (except by zero). It's the "Ultimate Rule Book."

Problem Solving: Proving a Group

Problem: Prove the set of Integers $\mathbb{Z}$ is a group under Addition $(+)$.

How to Solve (Check 4 rules):

1. **Closure:** If you add two integers, is the result an integer? $1 + 2 = 3$. Yes.
2. **Associativity:** Is $(a + b) + c = a + (b + c)$? Yes, addition is always associative.